1. Results

This chapter will present the results from the analysis on the dataset.

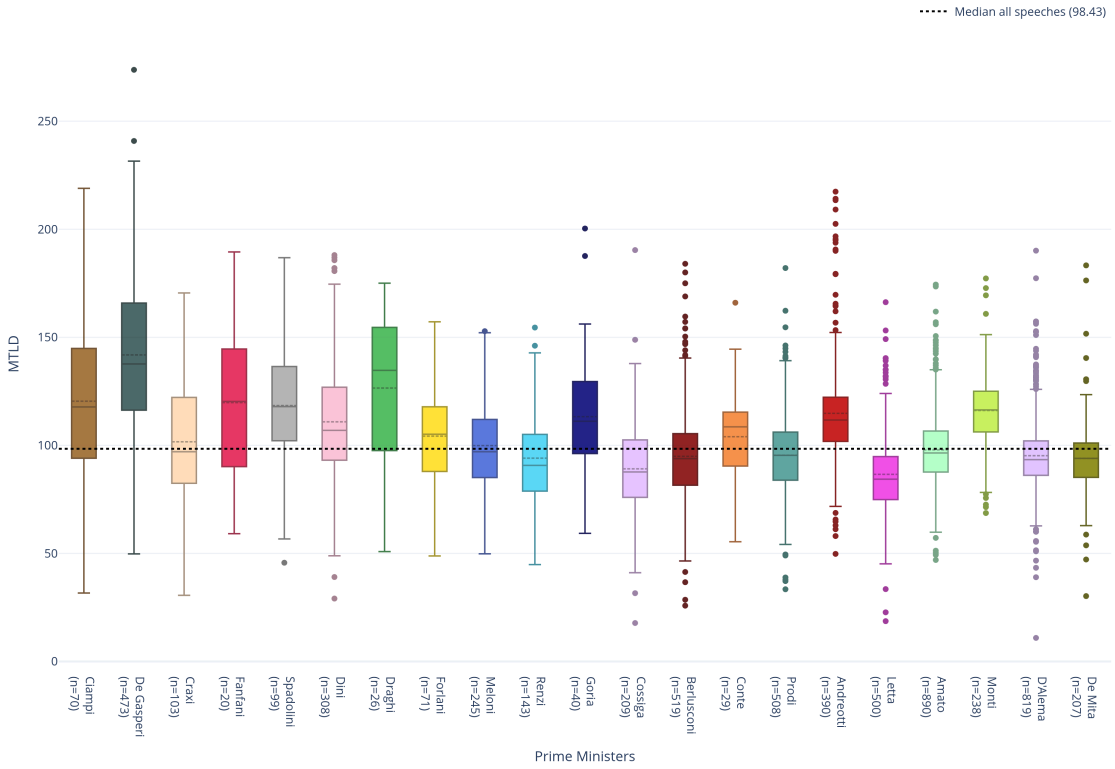
1.1. Textual Complexity

1.1.1. Prime Ministers' MTLDs

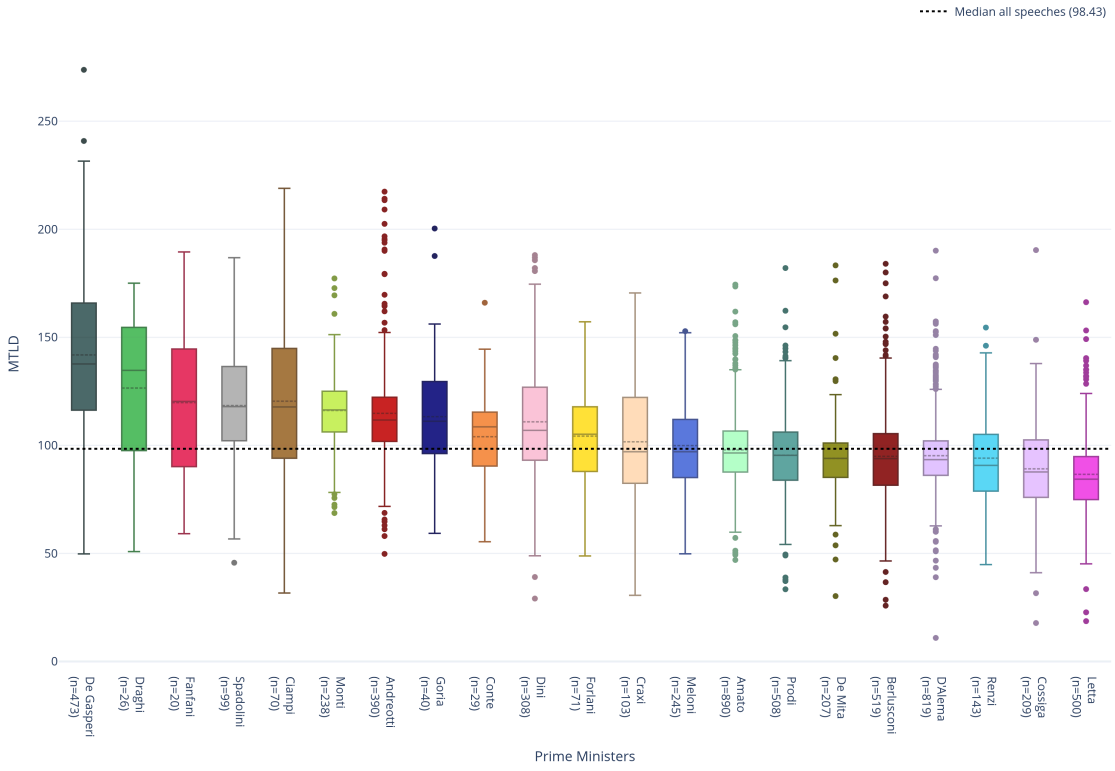
Let's first take a look at the Prime Ministers themselves. Figure Figure 1 shows two box plots where each box is a Prime Minister with at least twenty speeches, ordered by variability (whisker-to-whisker length) and median lexical diversity (median MTLD) respectively. We can visually divide the Prime Ministers into two groups: high- and low-variance. The first group, with seven out of twenty-one Prime Ministers (Ciampi, De Gasperi, Craxi, Fanfani, Spadolini, Dini, and Draghi) exhibits high variability in the complexity of their speeches, suggesting diverse speech styles. This is corroborated by their standard deviations (refer to table Table 1 for the exact values), all above the average of 24.32 (Ciampi, De Gasperi, Fanfani, and Draghi are even above thirty). Among these high-variance Prime Ministers there is a significant gap in variability: between Ciampi and Dini, for example, the interquartile range differs by 32.98%, going from 50.3 to 33.71.

The other Prime Ministers (pertaining to the low-variance group), seem more consistent in the complexity of their speeches. Instead, between Forlani and De Mita the interquartile range almost cuts in half, going from 28.76 to 15.86 for a reduction of 44.85%, telling us that in the low-variance group there are even stronger differences in variance between Prime Ministers. In this last group, two Prime Ministers have a standard deviation above average: Conte, whose standard deviation of 24.32 sits exactly on the average, and Goria with a standard deviation of 29.26. A possible explanation could be found in their low speech count, meaning the standard deviation is simply a noisy estimate. Incidentally, Fanfani and Draghi are in a similar position with twenty and twenty-six speeches each, and already sit in the high-variance group. De Gasperi's complexity stands above the rest even with a rich corpus of over 400 speeches. This could be explained either by a stylistic difference or by the fact that their speeches are the only ones not transcribed by Whisper

Overall, Prime Ministers in this study exhibit variation between one another, suggesting that there is not a "complexity typical of a Prime Minister", and that each one has their own style and richness of vocabulary.



(a) Box plot of MTLD values for each Italian Prime Minister, ordered by whisker-to-whisker length.



(b) Box plot of MTLD values for each Italian Prime Minister, ordered by median MTLD.

Figure 1: Figure 1a and Figure 1b show MTLD values for each Italian Prime Minister. The mean is a dotted line. Prime Ministers with fewer than 20 speeches are omitted.

Figure Figure 2 shows the evolution of the lexical complexity over each Prime Minister's career, via

LOWESS regression[1]. Once again, there does not seem to be any clear trend among Prime Ministers, a fact that is consistent with the last column of table Table 1, which shows the evolution of the MTLD (in percentage) during the span of the Prime Ministers' careers. Twelve Prime Ministers have positive changes, nine negative, so there is no common direction. Some show little to no change, like Andreotti (+2.18), Forlani (+1.51), and Monti (-2.89), while others change considerably, for example De Gasperi (-19.58), Cossiga (+23.64), Craxi (-23.75), and Meloni (+32.31). The MTLD evolution was calculated by fitting a linear regression to each Prime Minister's career, and then extracting the rescaled slope, where *Slope* is the fitted change in MTLD per percentage point of career elapsed, and *Intercept* is the fitted MTLD at the start of the career:

$$\text{Change } \% = \frac{\text{Slope} \cdot 100}{\text{Intercept}} \cdot 100 \qquad 1.$$

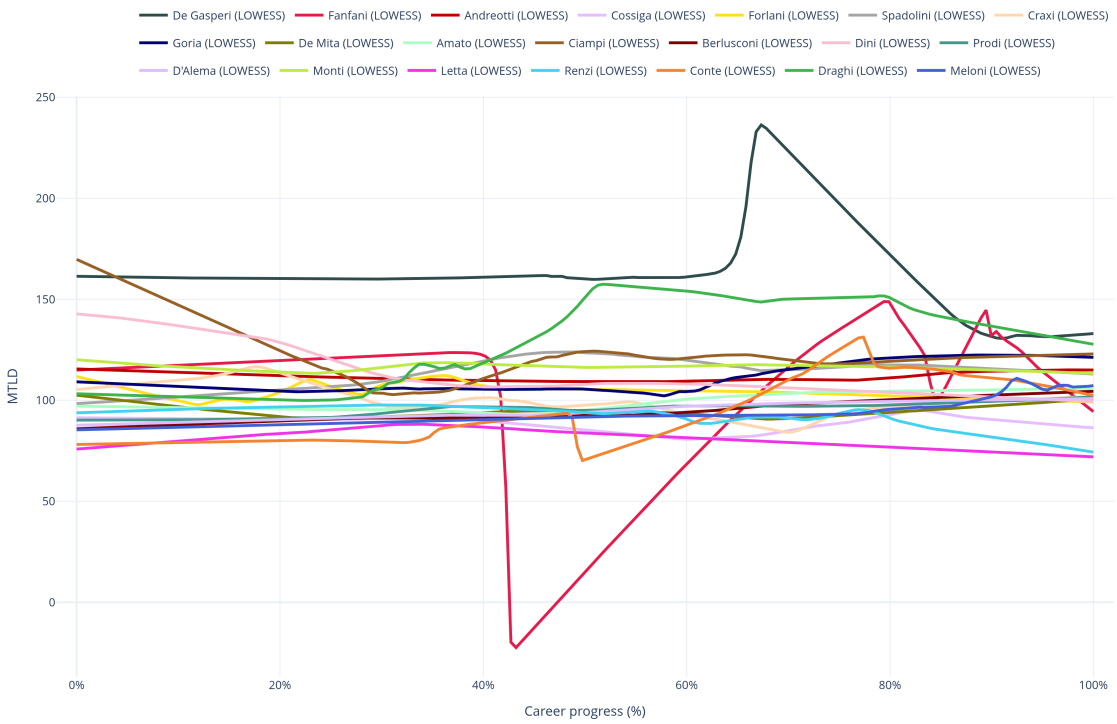
I used a linear regression instead of a LOWESS to extract one easily interpretable number. One thing to note: the LOWESS regressions for Draghi, Conte, and Fanfani look less consistent, which is expected since their datasets are scarce.

PM	# Speeches	Mean	Median	Std. dev.	IQR	# Outliers	Career change (%)
De Gasperi	473	141.83	137.64	35.88	49.50	2	−19.58
Fanfani	20	119.80	120.31	37.59	50.09	0	−9.93
Andreotti	390	114.81	111.74	23.77	20.42	28	+2.18
Cossiga	209	89.12	87.69	21.03	26.41	4	+23.64
Forlani	71	104.28	105.13	23.38	28.76	0	+1.51
Spadolini	99	118.43	117.94	25.58	34.01	1	+3.04
Craxi	103	101.63	97.06	27.29	39.60	0	−23.75
Goria	40	113.30	111.16	29.26	31.68	2	+11.11
De Mita	207	93.97	93.97	16.62	15.86	10	−6.26
Amato	890	97.90	96.47	16.07	18.98	25	+15.83
Ciampi	70	120.44	117.78	37.15	50.30	0	−5.31
Berlusconi	519	94.92	93.87	21.17	23.83	19	+22.84
Dini	308	110.88	106.95	25.97	33.71	8	−19.29
Prodi	508	95.42	95.38	19.81	22.25	16	+4.66
D'Alema	819	95.23	93.39	16.08	15.93	52	+17.23
Monti	238	116.15	116.40	16.42	18.57	11	−2.89
Letta	500	86.59	84.34	17.37	19.87	18	−11.50
Renzi	143	94.06	90.74	21.15	26.05	2	−16.06
Conte	29	104.02	108.58	24.32	24.67	1	+48.66
Draghi	26	126.53	134.68	33.52	55.17	0	+43.91
Meloni	245	99.89	97.04	21.20	26.85	1	+32.31

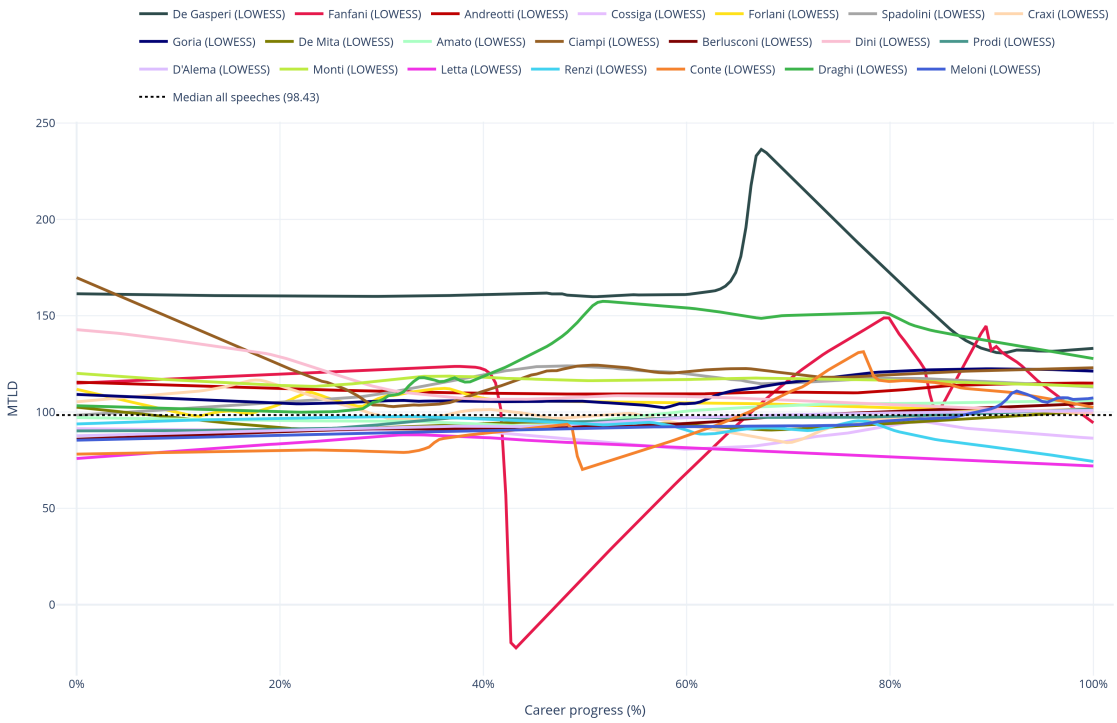
Table 1: Descriptive statistics of the MTLD (Measure of Textual Lexical Diversity) per Prime Minister, ordered by first term in office. In order: Prime Minister's name, number of speeches, mean,

[1]LOWESS was chosen for visualising the evolution of MTLD over linear models or splines because it combines much of the simplicity of linear least squares regression with the flexibility of nonlinear regression. In short, it is flexible but interpretable.

median, standard deviation, interquartile range (IQR), number of outliers (speeches outside the whiskers of the boxplot), and the MTLD evolution over their careers, calculated by fitting a linear regression over each Prime Minister's career. The percentage is relative to the fitted starting MTLD. Only Prime Ministers with at least twenty speeches are included.



(a) LOWESS curve of MTLD values for each Italian Prime Minister, without overall median.



(b) LOWESS curve of MTLD values for each Italian Prime Minister, with overall median.

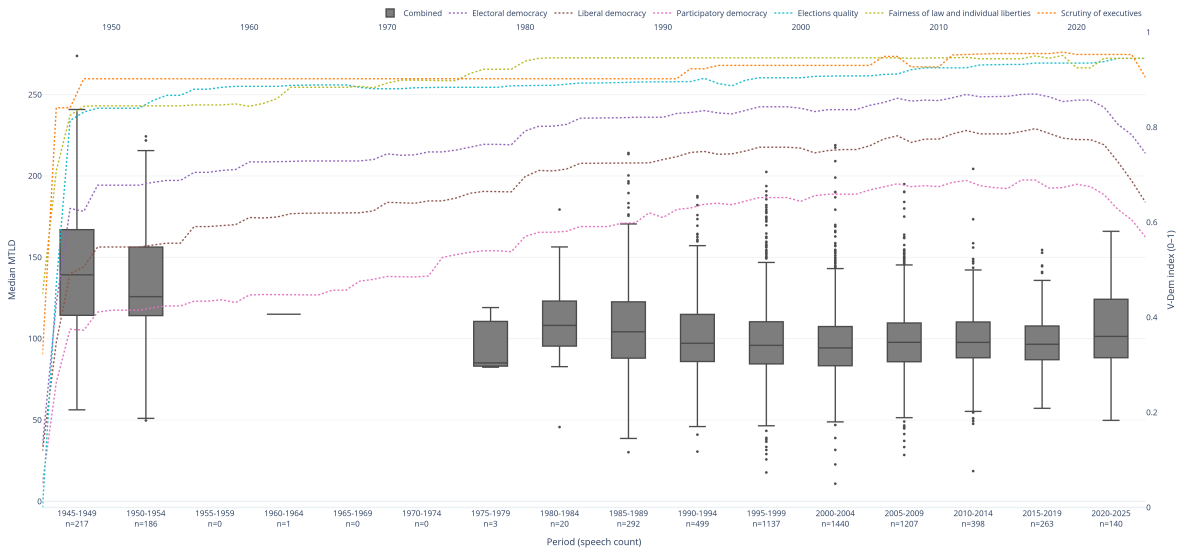
Figure 2: Figure 2a and Figure 2b show MTLD values for each Italian Prime Minister over their career span. The LOWESS curve was chosen to aid in the visual interpretation of the values. Prime Minister with fewer than 20 speeches are omitted.

1.1.2. VDEM

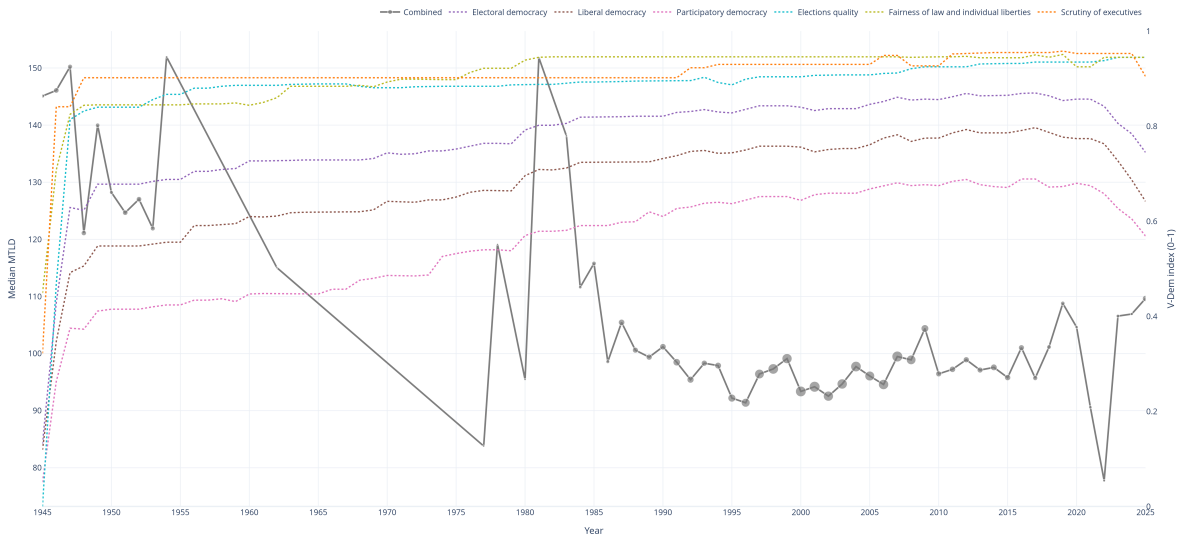
1.1.2.1. 5 years

The graphs at figure Figure 3 show the lexical diversity (MTLD) of all speeches by all Prime Ministers (without distinguishing between political leanings). The box plot has boxes aggregated by five year periods, while the line chart has each point has its size scaled by the amount of speeches present for that specific year. Measuring the MTLD separated into three distinct political leanings (left, center, and right) is not as informative due to the unequal distribution of speeches, as previously shown in figure TODOFIGURA, but the graphs with such aggregations are still viewable at appendix Figure 12.

Ignoring the period between 1960 and 1985 due to lack of data (only twenty speeches were collected in this year gap), the graphs shows a clear reduction in the variance of the MTLD across the board, until 2021 where it grow back again. This is corroborated by the data shown in table Table 2: interquartile range steadily decreases from 52.61, to 20.62, to grow back up to 35.37, while whisker-to-whisker length goes from 184.56, to 78.60, and grows back up to 116.21. While in the decreasing phase, both IQR and whisker-to-whisker length decrease steadily for each 5-year period. In the graph, democratic indices are also reported…



(a) MTLD values in a Boxplot by 5-year periods, with VDEM indices. Each grey box combines all speeches for all political leanings in that 5-year period.



(b) MTLD values plotted for each year, aggregating all speeches from all political leanings. Each dot of the graph (for each year) is scaled based on the amount of speeches available for that specific year. The MTLD line chart is plotted against VDEM indices.

Figure 3: MTLD values in a Boxplot by 5-year periods. There seem to be a reduction in MTLD variance starting from 1989 all the way to 2019, whith an abrupt inversion in the 2020-2025 period. If this is a correlation with the inflection of some VDEM indices, it is not clear.

Period	# Speeches	Median MTLD	IQR	Whisker-to-whisker	# Outliers
1945-1949	217	139.24	52.61	184.56	1
1950-1954	186	125.82	41.72	155.95	5
1960-1964	1	115.04	0.00	0.00	0
1975-1979	3	85.14	18.32	36.65	0
1980-1984	20	108.15	27.04	73.58	2

1985-1989	292	104.24	34.53	131.82	12
1990-1994	499	97.20	28.87	111.17	16
1995-1999	1137	95.91	25.86	100.37	54
2000-2004	1440	94.33	24.08	94.23	40
2005-2009	1207	97.78	23.79	93.89	41
2010-2014	398	97.75	21.88	85.88	16
2015-2019	263	96.56	20.62	78.60	6
2020-2025	140	101.42	35.37	116.21	0

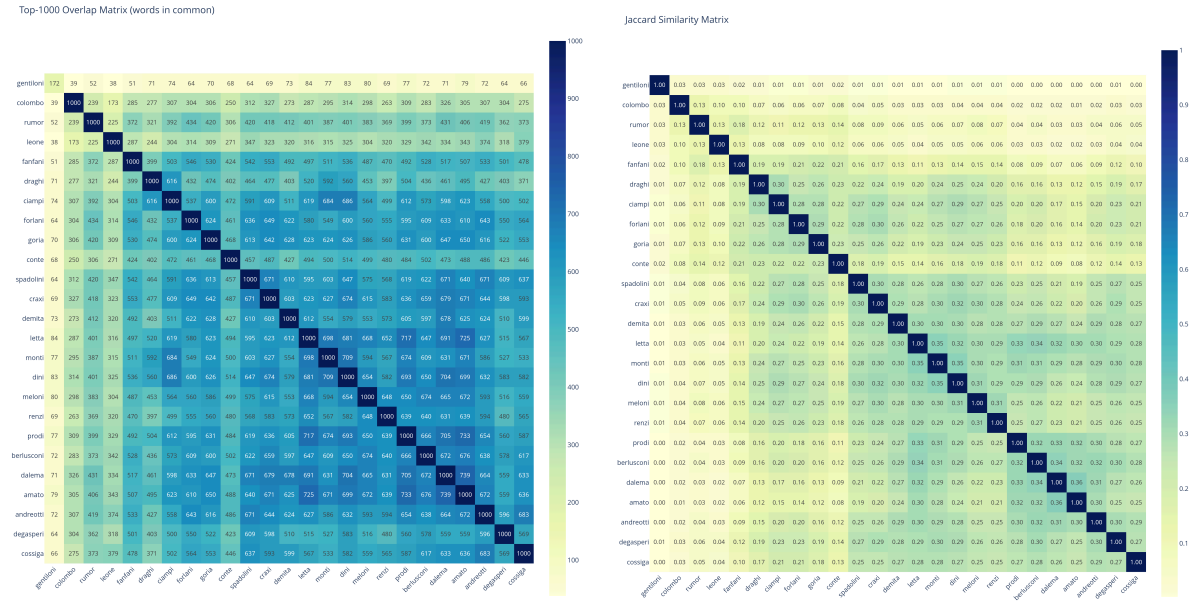
Table 2: MTLD (Measure of Textual Lexical Diversity) per five-year period, with the speeches of all prime ministers pooled. In order: period, number of speeches, median MTLD, interquartile range (IQR), distance between the lower and upper whisker of the boxplot, and number of outliers (speeches outside the whiskers). Periods with no speeches are not shown.

1.1.3. Conclusions

There does not seem to be a clear correlation between MTLD values and democratic indices, with the exception for the variance which ...

1.2. Polarization

1.2.1. Prime Minister's Lexical Similarity

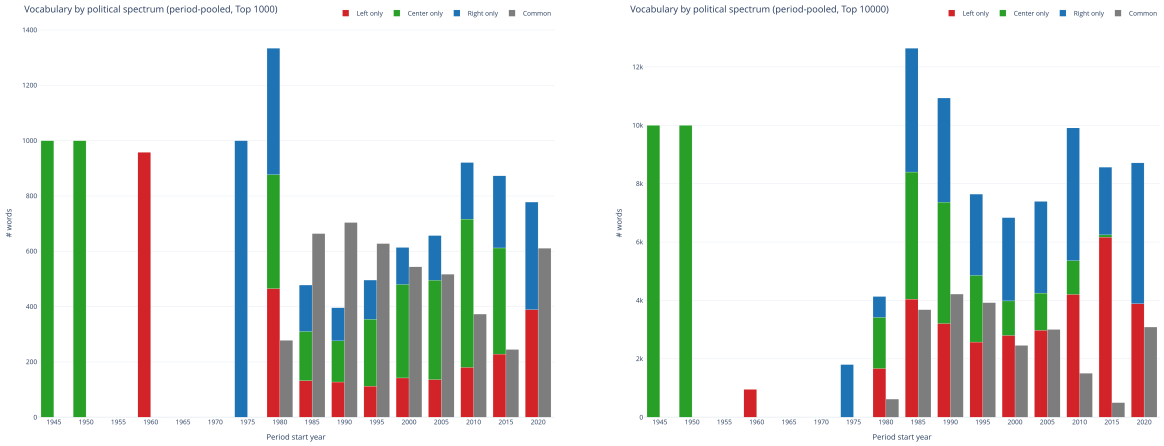


(a) Words similarity.

(b) Jaccard similarity.

Figure 4: Heatmaps showing lexical similarity between Prime Ministers' speeches. Figure 4a was made by pooling the 1000 most used words by each Prime Minister, then calculating how many words are shared between each of them. Figure 4b, on the other hand, is made by calculating the Jaccard index between each pair of Prime Ministers, by pooling *all* words from all speeches used by each Prime Minister.

1.2.2. Political Leanings Lexicons Over Time



(a) Top 1,000 pooling.

(b) Top 10,000 pooling

Figure 5: All speeches are pooled in 5-year time frames. Then, for each 5-year time frame, the top-N most frequently used words are extracted for each political leaning[2] (one thousands words for graph at Figure 5a, and ten thousands words for graph at Figure 5b). This results in three distinct sets[3] of words for left, right, and center leaning speeches respectively.

The graph shows, for each 5-year time frame, words in common among all leanings (in grey), and words unique for each leaning (in red, blue, and green respectively). For the years with most data (from 1985 to 2015), Figure 5a shows a progressive increase in unique vocabularies compared to a decrease in shared ones, which can be interpreted as symptoms of increasing polarization.

[2] A speech is considered "right-leaning" if it pertains a right-leaning Prime Minister.

[3] When speeches for each leaning are present. Otherwise only two, one, or no set at all can be extracted from that 5-year time frame.

1.2.3. t-SNE

1.2.3.1. Word2Vec: Word Embeddings

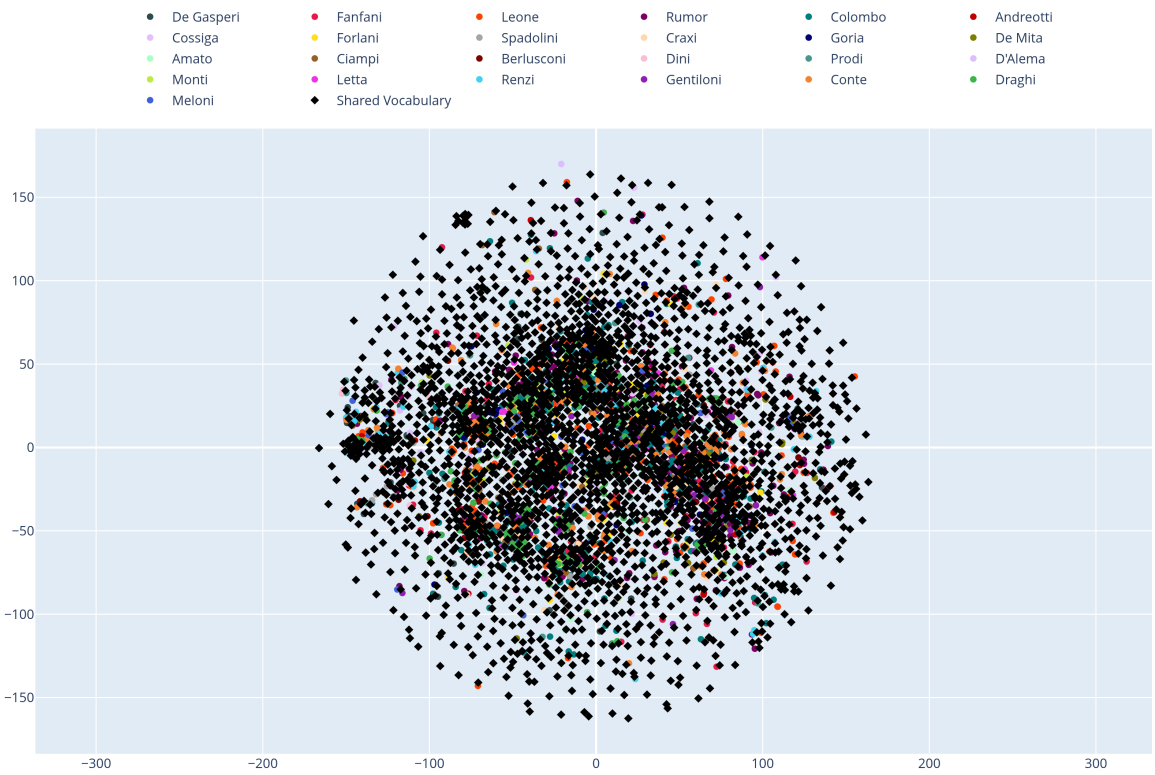
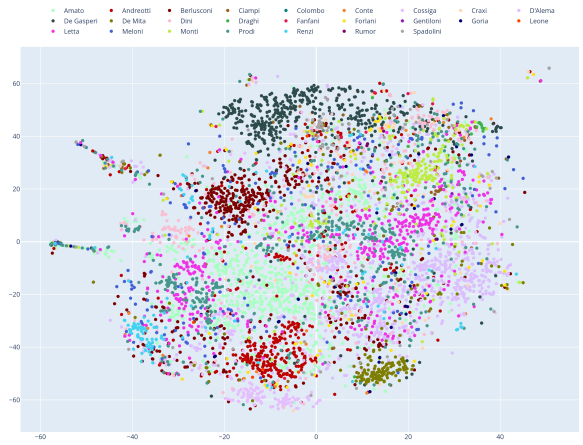


Figure 6: Top-1000 words[4] from each Prime Minister. The embedding vectors are pretrained Italian word vectors (CBOW, 300 dimensions) taken from *fastText*. Words shared by all Prime Ministers are displayed as black rhombuses. The graph show a lot of clustering, suggesting a certain similarity among Prime Ministers. That being said, the graph is also too hard to read to gain any conclusive evidence.

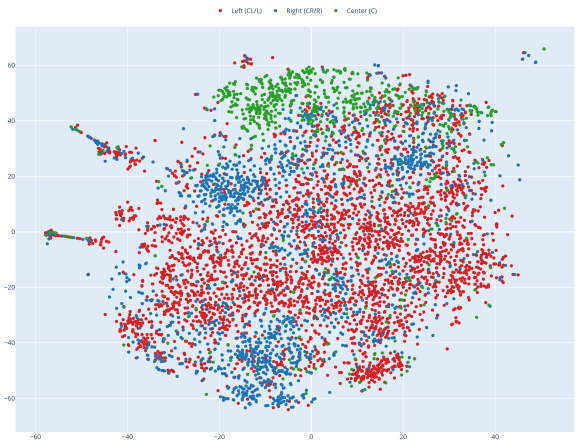
,

[4]Top-1000 words means pooling the one thousand most used words of a specific Prime Minister. Stopwords are removed and the lexicon is lemmatized.

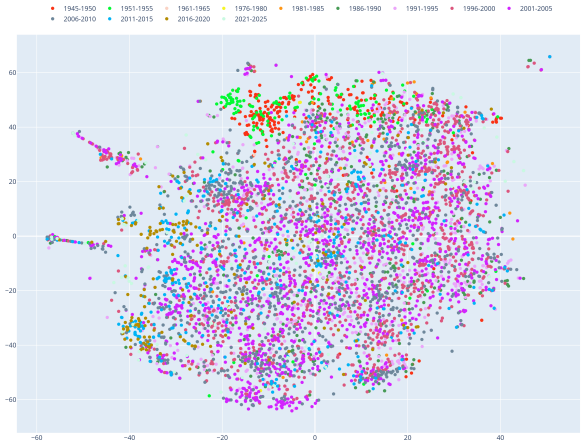
1.2.3.2. Doc2Vec: Speech Embeddings



(a) Doc2Vec: Prime Ministers



(b) Doc2Vec: political leaning



(c) Doc2Vec: year periods

Figure 7: Using `TfidfVectorizer` from `sklearn` library, I computed one embedding for each speech[5]. The embeddings can then be projected in two dimensions. Figure 7a shows the projected speech vectors grouped by Prime Ministers. We can see clustering, suggesting that most Prime Ministers have a lexical style and wording that are unique to them. Figure 7b groups the vectors by political leaning, showing that different leanings occupy different and distinct regions in the vector space. Both graphs strongly hints to a polarized political landscape.

On the other hand, Figure 7c shows two main clustering regions: on the top, speeches from 1945 to 1955 gather, while speeches from 1991 to 2025 are more uniformly scattered across the whole vector space. This suggests that earlier speeches were significantly different compared to modern ones, while speeches from the nineties onward are quite similar.

TODO: compare with VDEM polarization indices (if exists).

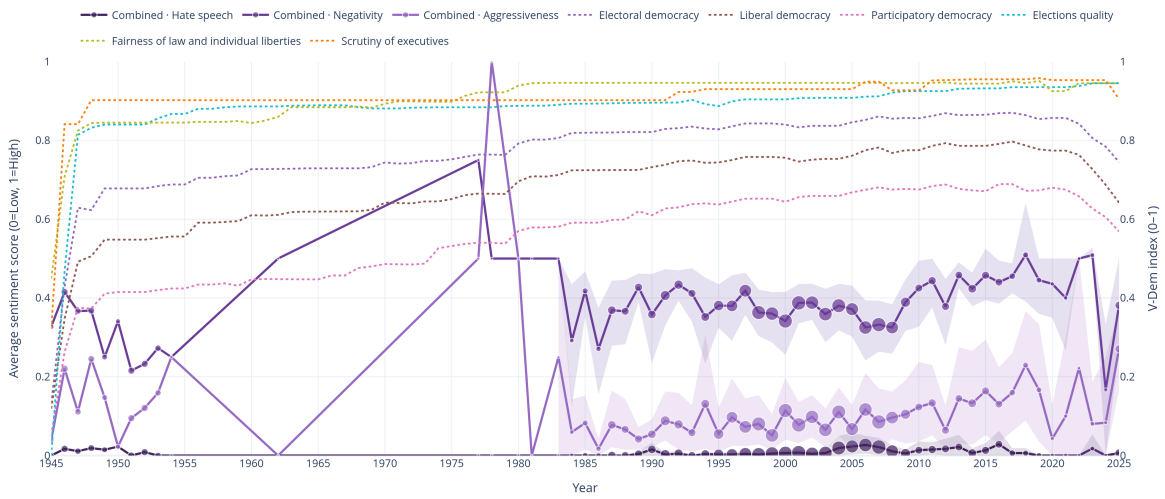
1.2.4. Conclusions

We see quite a bit of polarization, but ...

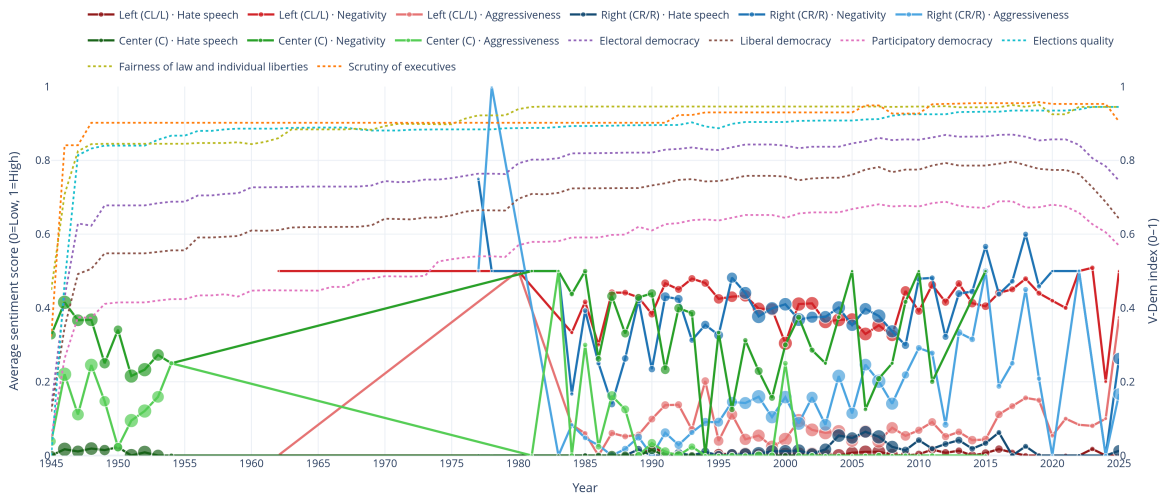
[5]Each speech counts as a document.

1.3. Sentiment Analysis

1.3.1. Democratic Indices Against Sentiment Scoring



(a) Sentiment values plotted for each year, aggregating all speeches from all political leanings. Each dot of the graph (for each year) is scaled based on the amount of speeches available for that specific year. The confidence intervals are shown. The sentiment scores chart is plotted against VDEM indices.

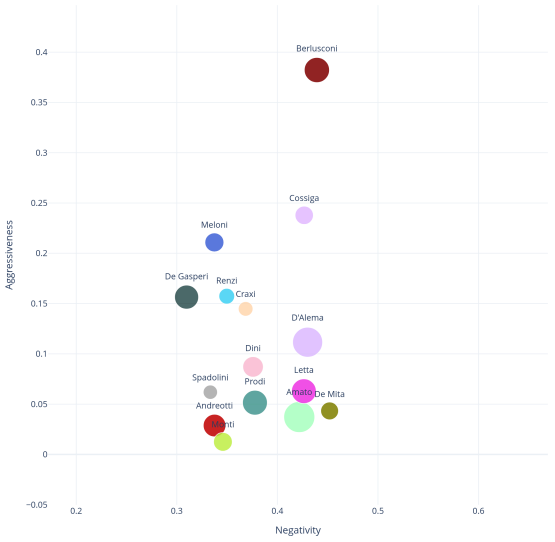


(b) Sentiment values plotted for each year, aggregating speeches from three political leanings, respectively colored as green (center), red (left), and blue (right). Each dot of the graph (for each year) is scaled based on the amount of speeches available for that specific year. The The sentiment scores line chart is plotted against VDEM indices.

Figure 8: Sentiment analysis results for *hate speech*, *aggressiveness*, and *negativity* plotted against VDEM indices. Its hart to tell wether there is any correlation between them.

1.3.2. Toxicity (TODO: change name)

First thing first: hate speech is so low for everyone that we can ignore it



(a) Sentiment: Prime Ministers ($|\text{speeches}| \geq 100$).



(b) Sentiment: leanings.



(c) Sentiment: years zoomed out.



(d) Sentiment: years zoomed in.

Figure 9: Sentiment analysis results for *negativity* and *aggressiveness*. Values go from zero to one, and the classification task considers three possibile values for each speech: *low* (0), *mid* (0.5), and *high* (1).

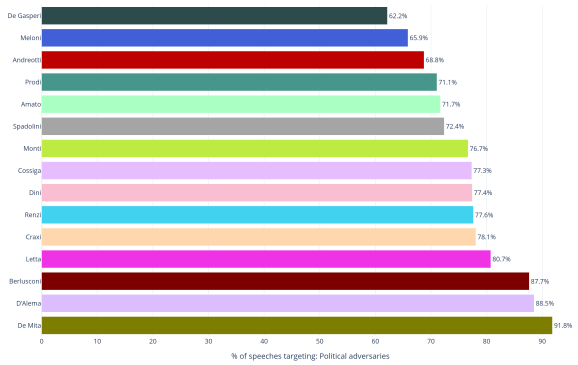
Size of bubbles depends on the amount of speeches. Figure 9a project results for single Prime Ministers: Berlusconi stands out as the most aggressive by far. Figure 9b project results aggregated by leanings. The right appears to be more aggressive, while the left more negative. The center seems to be more positive than both. The last two figures show the sentiment scores aggregated by years (with 5-year pool range). While Figure 9c shows an outlier for the period 1976 — 1980, Figure 9d shows that overall the sentiment scores are quite close to each other, without great variance especially for periods with a lot of speeches.

1.3.3. Targets

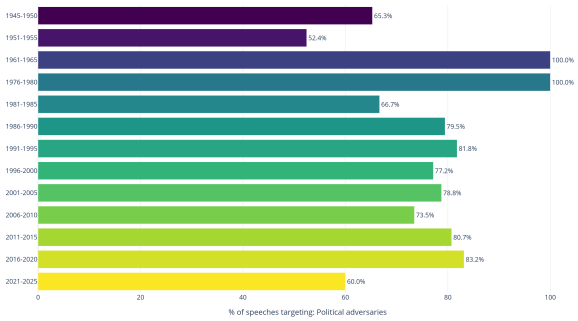
Five possible targets:

- None;
- Political adversaries;
- Gender minorities;
- Religious minorities;
- Ethnic minorities.

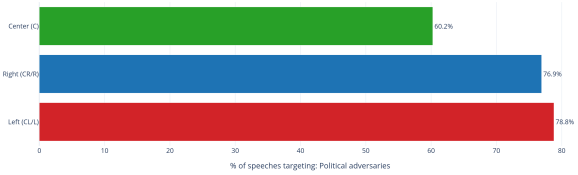
The targets are classified independently to the sentiment scores. For example, a speech could have *low* aggressiveness and still be targeting political adversaries.



(a) Target: political adversaries. Grouped by: PMs.



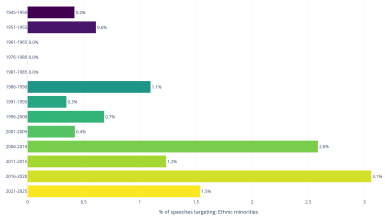
(b) Target: political adversaries. Grouped by: years.



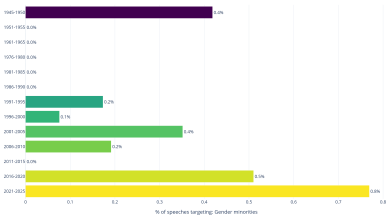
(c) Target: political adversaries. Grouped by: leanings.

Figure 10: All graphs show the percentage of speeches that target political adversaries. Figure 10a shows which Prime Minister speaks more often about the opposition (Prime Ministers with less than 100 speeches are omitted). Figure 10b shows in which 5-year periods there are more speeches targeting political adversaries (ordered by year). Lastly, Figure 10c shows which of the three political fields is more likely to speak about the opposing front. The results show how, across the board, more than one speech out of two targets the opposition. This is true regardless of time period or affiliation.

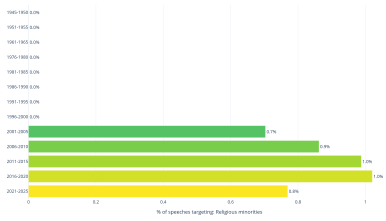
Thankfully, Italian Prime Minister apparently do not target minorities often.



(a) Target: ethnic minorities.



(b) Target: gender minorities.



(c) Target: religious minorities.

Figure 11: Overall, an extremely low percentage of speeches target minority groups. Therefore, I will ignore this analysis. (TODO explain better.. within error yada yada).

1.3.4. Conclusions

Italian Prime Minister seem to be overall a bit aggressive and negative, and they speak more often then not about the opposition. Almost no traces of hate speech and attacks against minority groups are found in the data.

This does not seem to change over time, failing to explain the VDEM indices drop in 2021.

References

2. Discussion & Future Works

- Analyses results
  - so can we explain VDEM drops and stuff?
- How to improve the work
  - more scraping (increase dataset)
  - better transcription (even tho they are pretty good)
  - bigger LLMs for the sentiment analysis
  - better doc2vec? gensim still not compatible with current Python version
- How to expand the work
  - include leader of opposition
  - include trend setter of the time
  - include other countries (eg France or USA)

A Appendix

1.1. Radio Radicale HTML List of Prime Minister's Speeches

```
  <li class="views-row views-row-1 clearfix">
        <div class="tipo_media audio">audio</div>

        <div class="ls_text">
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datatype="xsd:dateTime" content="1993-07-25T00:00:00+02:00">25.07.1993</
span>
        <h3><a href="/scheda/55884/per-litalia-una-nuova-presenza-popolare?
i=2447650">"Per l'Italia una nuova presenza popolare"</a></h3>


        </div>
        <div class="ls_meta">
        <p class="dove"><span class="date-display-single" property="dc:date"
datatype="xsd:dateTime" content="1993-07-25T02:39:01+02:00">2:39</span></p>
        <p class="durata">Durata: 0:05:00</p>
        </div>  </li>
<li class="views-row views-row-2 clearfix">
        <div class="tipo_media audio">audio</div>

        <div class="ls_text">
        <span class="date-display-single" property="dc:date"
datatype="xsd:dateTime" content="1992-04-15T00:00:00+02:00">15.04.1992</
span>
        <h3><a href="/scheda/46078/consiglio-nazionale-della-dc?
i=2529451">Consiglio Nazionale della DC</a></h3>


        </div>
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datatype="xsd:dateTime" content="1992-04-15T02:29:01+02:00">2:29</span></p>
        <p class="durata">Durata: 0:04:00</p>
        </div>  </li>
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datatype="xsd:dateTime" content="1991-11-30T00:00:00+01:00">30.11.1991</
span>
        <h3><a href="/scheda/43012/conferenza-nazionale-della-democrazia-
cristiana?i=2554258">Conferenza Nazionale della Democrazia Cristiana</a></
h3>


        </div>
        <div class="ls_meta">
        <p class="dove"><span class="date-display-single" property="dc:date"
datatype="xsd:dateTime" content="1991-11-30T07:15:02+01:00">7:15</span></p>
        <p class="durata">Durata: 0:11:00</p>
        </div>  </li>
<li class="views-row views-row-4 clearfix">
        <div class="tipo_media audio">audio</div>

        <div class="ls_text">
```

```
        <span class="date-display-single" property="dc:date"
datatype="xsd:dateTime" content="1991-09-28T00:00:00+02:00">28.09.1991</
span>
        <h3><a href="/scheda/41372/riforme-istituzionali-e-sistema-dei-partiti?
i=2567492">"Riforme istituzionali e sistema dei partiti"</a></h3>


        </div>
        <div class="ls_meta">
        <p class="dove"><span class="date-display-single" property="dc:date"
datatype="xsd:dateTime" content="1991-09-28T02:32:01+02:00">2:32</span></p>
        <p class="durata">Durata: 0:10:00</p>
        </div>  </li>
<li class="views-row views-row-5 clearfix">
        <div class="tipo_media audio">audio</div>

        <div class="ls_text">
        <span class="date-display-single" property="dc:date"
datatype="xsd:dateTime" content="1989-11-18T00:00:00+01:00">18.11.1989</
span>
        <h3><a href="/scheda/33836/consiglio-nazionale-dc?i=2631437">Consiglio
nazionale Dc</a></h3>


        </div>
        <div class="ls_meta">
        <p class="dove"><span class="date-display-single" property="dc:date"
datatype="xsd:dateTime" content="1989-11-18T03:49:02+01:00">3:49</span></p>
        <p class="durata">Durata: 0:13:00</p>
        </div>  </li>
<li class="views-row views-row-6 clearfix">
        <div class="tipo_media audio">audio</div>

        <div class="ls_text">
        <span class="date-display-single" property="dc:date"
datatype="xsd:dateTime" content="1989-09-06T00:00:00+02:00">6.09.1989</span>
        <h3><a href="/scheda/58780/lo-schok-da-perestrojka?i=2419587">"Lo schok
da perestrojka"</a></h3>


        </div>
        <div class="ls_meta">
        <p class="durata">Durata: 0:00:01</p>
        </div>  </li>
<li class="views-row views-row-7 clearfix">
        <div class="tipo_media audio">audio</div>

        <div class="ls_text">
        <span class="date-display-single" property="dc:date"
datatype="xsd:dateTime" content="1989-09-06T00:00:00+02:00">6.09.1989</span>
        <h3><a href="/scheda/58780/lo-schok-da-perestrojka?i=2419589">"Lo schok
da perestrojka"</a></h3>


        </div>
        <div class="ls_meta">
        <p class="durata">Durata: 0:00:01</p>
        </div>  </li>
<li class="views-row views-row-8 clearfix">
        <div class="tipo_media audio">audio</div>
```

```
92
93          <div class="ls_text">
            <span class="date-display-single" property="dc:date"
94  datatype="xsd:dateTime" content="1989-09-06T00:00:00+02:00">6.09.1989</span>
            <h3><a href="/scheda/58780/lo-schok-da-perestrojka?i=2419592">"Lo schok
95  da perestrojka"</a></h3>
96
97    ................
98
99          </div>
100         <div class="ls_meta">
101         <p class="durata">Durata: 3:00:00</p>
102         </div>  </li>
```

Listing 1: Example of the HTML in Radio Radicale that contains a list of links pointing to the speeches of a Prime Minister.

1.2. Radio Radicale HTML List of Interventions

```
1               <div class="l-region l-region--seconda-colonna">
      <div id="block-quicktabs-blocco-interventi" class="block block--quicktabs
2   block--quicktabs-blocco-interventi block-first block-1 block-last">
3     <div class="block__content">
                  <div id="quicktabs-blocco_interventi" class="quicktabs-wrapper
    quicktabs-style-nostyle jquery-once-1-processed"><ul class="quicktabs-tabs
    quicktabs-style-nostyle"><li class="active"><a href="/scheda/41372/riforme-
    istituzionali-e-sistema-dei-partiti?i=2567492&amp;qt-blocco_interventi=0#qt-
4   blocco_interventi" id="quicktabs-tab-blocco_interventi-0" class="quicktabs-tab
    quicktabs-tab-block quicktabs-tab-block-views-
    delta-87664a805c1e84a93a9afe91f2d174cd active quicktabs-loaded jquery-once-2-
    processed">Interventi<span id="active-quicktabs-tab" class="element-
    invisible">(scheda attiva)</span></a></li>
    <li><a href="/scheda/41372/riforme-istituzionali-e-sistema-dei-partiti?
    i=2567492&amp;qt-blocco_interventi=1#qt-blocco_interventi" id="quicktabs-tab-
5   blocco_interventi-1" class="quicktabs-tab quicktabs-tab-block quicktabs-tab-
    block-rrblocks-delta-box-trascrizione active jquery-once-2-
    processed">Trascrizione automatica</a></li>
    </ul><div id="quicktabs-container-blocco_interventi" class="quicktabs_main
    quicktabs-style-nostyle"><div id="quicktabs-tabpage-blocco_interventi-0"
6   class="quicktabs-tabpage "><div id="block-
    views-87664a805c1e84a93a9afe91f2d174cd" class="block block--views block--
    views-87664a805c1e84a93a9afe91f2d174cd block-first block-1">
7     <div class="block__content">
                  <div class="view view-interventi-2016 view-id-interventi_2016
8   view-display-id-blocco_interventi view-dom-id-cdc25d3e2de2a31806b2fe74f563f3ef">
9
10
11
12        <div class="view-content int_menu">
13        <ul>  <li class="int2567502 intervento d2220">
14
15
16            <div class="int_name_section">
17            <h2>Flaminio Piccoli</h2><p>DC</p>
18            </div>
19            <div class="int_subtext">
20            </div>
21            <div class="durata"> 0:00 Durata: 37 min</div>
22                <div class="actions" style="display: none;">
```

```
                <a href="/soggetti/247/flaminio-piccoli" title="altri interventi
dell'oratore"><i class="fa fa-external-link" aria-hidden="true"></i>altri
interventi</a>
          <a class="condividi condividi-intervento" href="#" data-param-
i="2567502"><i class="fa fa-share-alt"></i>condividi intervento</a>
    </div>

</li>
  <li class="int2567501 intervento d1800">


          <div class="int_name_section">
          <h2>Carlo Bernini</h2><p>MINISTRO</p>
          </div>
          <div class="int_subtext">
          </div>
          <div class="durata"> 0:37 Durata: 30 min</div>
              <div class="actions" style="display: none;">
              <a href="/soggetti/19978/carlo-bernini" title="altri interventi
dell'oratore"><i class="fa fa-external-link" aria-hidden="true"></i>altri
interventi</a>
          <a class="condividi condividi-intervento" href="#" data-param-
i="2567501"><i class="fa fa-share-alt"></i>condividi intervento</a>
    </div>

</li>
  <li class="int2567500 intervento d960 over">


          <div class="int_name_section">
          <h2>Lucio Toth</h2><p>DC</p>
          </div>
          <div class="int_subtext">
          </div>
          <div class="durata"> 1:07 Durata: 16 min</div>
              <div class="actions" style="display: block;">
              <a href="/soggetti/16113/lucio-toth" title="altri interventi
dell'oratore"><i class="fa fa-external-link" aria-hidden="true"></i>altri
interventi</a>
          <a class="condividi condividi-intervento" href="#" data-param-
i="2567500"><i class="fa fa-share-alt"></i>condividi intervento</a>
    </div>

..................

</li>
  <li class="int2567473 intervento d1800 ">


          <div class="int_name_section">
          <h2>Rosa Iervolino Russo</h2><p>MINISTRO</p>
          </div>
          <div class="int_subtext">
          </div>
          <div class="durata"> 6:56 Durata: 30 min</div>
              <div class="actions" style="display:none;">
              <a href="/soggetti/25029/rosa-iervolino-russo" title="altri
interventi dell'oratore"><i class="fa fa-external-link" aria-hidden="true"></
i>altri interventi</a>
```

```
            <a class="condividi condividi-intervento" href="#" data-param-
i="2567473"><i class="fa fa-share-alt"></i>condividi intervento</a>
    </div>

</li>
</ul>    </div>
```

Listing 2: Example of the HTML in Radio Radicale that contains a list of interventions in a conference.

1.3. LLMs Model Comparison for Sentiment Analysis

Model	Thinking	Label	κ	Accuracy	Macro Precision	Macro Recall	Macro F1
qwen2.5:7b-instruct-q4_K_M	-	Hate speech	0.173	0.696	0.634	0.622	0.584
qwen2.5:7b-instruct-q4_K_M	-	Negativity	0.297	0.570	0.579	0.612	0.566
qwen2.5:7b-instruct-q4_K_M	-	Aggressiveness	0.255	0.623	0.682	0.676	0.594
qwen2.5:7b-instruct-q4_K_M	-	Target	0.705	0.773	0.807	0.865	0.824
mistral:7b-instruct-v0.3-q4_K_M	-	Hate speech	0.128	0.729	0.650	0.682	0.616
mistral:7b-instruct-v0.3-q4_K_M	-	Negativity	0.061	0.382	0.326	0.421	0.348
mistral:7b-instruct-v0.3-q4_K_M	-	Aggressiveness	0.051	0.546	0.538	0.613	0.499
mistral:7b-instruct-v0.3-q4_K_M	-	Target	0.699	0.768	0.868	0.873	0.848
qwen3.5:4b-q8_0	off	Hate speech	0.091	0.560	0.463	0.497	0.475
qwen3.5:4b-q8_0	off	Negativity	0.222	0.502	0.620	0.494	0.451
qwen3.5:4b-q8_0	off	Aggressiveness	0.194	0.614	0.689	0.666	0.599
qwen3.5:4b-q8_0	off	Target	0.783	0.836	0.890	0.906	0.891
qwen3.5:4b	off	Hate speech	0.268	0.628	0.586	0.585	0.581
qwen3.5:4b	off	Negativity	0.208	0.459	0.424	0.446	0.397
qwen3.5:4b	off	Aggressiveness	0.363	0.691	0.736	0.736	0.683
qwen3.5:4b	off	Target	0.760	0.816	0.886	0.898	0.879
qwen3.5:9b-q4_K_M	off	Hate speech	0.037	0.517	0.501	0.486	0.476
qwen3.5:9b-q4_K_M	off	Negativity	0.246	0.522	0.599	0.518	0.483
qwen3.5:9b-q4_K_M	off	Aggressiveness	−0.009	0.541	0.535	0.560	0.538
qwen3.5:9b-q4_K_M	off	Target	0.721	0.787	0.872	0.878	0.864
gemma4:e4b	off	Hate speech	0.168	0.739	0.683	0.662	0.666
gemma4:e4b	**off**	**Negativity**	**0.293**	**0.536**	**0.534**	**0.564**	**0.532**
gemma4:e4b	**off**	**Aggressiveness**	**0.476**	**0.691**	**0.701**	**0.726**	**0.690**
gemma4:e4b	off	Target	0.674	0.758	0.833	0.837	0.835
gemma4:e4b	**on**	**Hate speech**	**0.538**	**0.826**	**0.819**	**0.777**	**0.773**
gemma4:e4b	on	Negativity	0.139	0.546	0.604	0.577	0.561
gemma4:e4b	on	Aggressiveness	0.246	0.604	0.689	0.644	0.614

gemma4:e4b	**on**	**Target**	**0.869**	**0.903**	**0.942**	**0.927**	**0.930**
gemma4:12b*	off	Hate speech	0.228	0.743	0.691	0.656	0.664
gemma4:12b*	off	Negativity	0.230	0.563	0.583	0.625	0.552
gemma4:12b*	off	Aggressiveness	0.303	0.655	0.702	0.707	0.634
gemma4:12b*	off	Target	0.722	0.786	0.865	0.881	0.854

Table 3: Full validation results on the 207 speeches for all models, identified by their Ollama tags, with thinking mode off and on (- where thinking mode does not apply). In order: model, thinking mode, label, Cohen's κ (quadratic-weighted for hate speech, negativity and aggressiveness; unweighted for target), accuracy, and macro-averaged precision (P), recall (R) and F1. Rows in bold mark the configuration adopted for each label (`gemma4:e4b`). *Evaluated on 206/207 speeches, as one failed to be classified.

1.4. Prompt to Generate 100 Synthetic Speeches With High Hostility

```
look at this speech from giorgia meloni:
#-------begin of csv----------#

politician,historical_date,location,tags,description,title,url,audio_file,text,
hate_speech,negativity,aggressiveness,
target,hate_speech,negativity,aggressiveness,target
meloni,2026-05-04,EREVAN,[],"Vertice della Comunità Politica Europea  4 Maggio
2026  Il Presidente del Consiglio, Giorgia Meloni, ha partecipato oggi a
Yerevan, in Armenia, all'8° Vertice della Comunità Politica Europea (CPE), dove
ha tenuto un intervento al panel di discussione su “Maintaining European unity
and coherence in times of polycrisis”, moderato dal Presidente cipriota
Christodoulides. Prima di partire alla volta di Baku, il Presidente ha tenuto un
punto stampa. #yerevan #cpe #armenia #giorgiameloni  #governomeloni","Vertice
della Comunità Politica Europea, l'intervento del Presidente Giorgia
Meloni",https://youtu.be/6gJWd0_AlTI?si=QEgiswzWpwDsMzNn,crossdem/datasets/
meloni/audio_out/sMzNn_audio.mp3,"E sono completamente d'accordo con questo
approccio, che l'importanza strategica del Mediterraneo e del miglior Mio Oste,
promuovendo un approccio di 360 gradi tra l'estero e le flacche del sud, ma
anche a diversi tipo di problemi comuni. Uno dei problemi comuni che abbiamo è
la migrazione. In un ambiente molto complesso e internazionale oggi, come i
paesi europei, come l'Italia, potrebbero migliorare i prezzetti del
miglioramento? Abbiamo la crisi nel Mio Oste, quindi la migrazione è qui.
Inoltre, la migrazione, la stabilità e la sicurezza in Europa sono in contatto.
Grazie molto, Nikos. Buongiorno, amici, e grazie molto al Prime Minister
Vashinian per questa fantastica organizzazione. Siamo oggi parlando di una crisi
polica, ma cosa significa una crisi polica? Non è semplicemente che stiamo
affrontando molte crisi insieme, è tutto. Il problema è che tutte queste crisi
sono rilasciate ad uno all'altro e aiutate l'altro. E penso che sia molto
importante capire questo, anche per affrontare il fatto che non possiamo
semplicemente affrontare una o due facette di questa crisi, ma dobbiamo
affrontare tutte le facette. E io utilizzerò esattamente, per cui Nikos
chiedeva, la migrazione, l'esempio della migrazione. L'esempio della migrazione
è anche linkato all'energia, perché dobbiamo anche affrontare il fatto che molte
flotte vengono dalle regioni instable che sono la chiave dei nostri prodotti di
energia. Ultimamente, tutto questo effettua la qualità delle nostre democrazie.
Perché quando i cittadini pensano che le difficoltà importanti non sono
governate, si perdono la fiducia nelle istituzioni. E quando la fiducia è
erogata, le persone si diventano più vulnerabili alla manipolazione, inoltre
all'uso maliccioso dell'intelligenza artificiale. Quindi non è migrazione. È
migrazione, è economia, è democrazia, è competitività, è energia, è sicurezza. E
se prendo uno degli altri problemi, posso linkarli tutti in modo stesso. Quindi
prima di tutto non possiamo affrontare solo un pezzo di questa crisi polica.
Secondo, la migrazione è parte integrale della crisi polica. Terzo, nessuno può
```

```
affrontarlo solo. Quindi la cooperazione su questo è una precondizione. Abbiamo
fatto dei passaggi molto importanti negli ultimi anni, al livello dell'Unione
Europea, dove ora abbiamo un nuovo Pacto di Assilio, abbiamo una nuova
definizione dei terzi paesi sicuri, stiamo andando verso una nuova regolazione
sui risultati che ci aiuterà anche su, diciamo, soluzioni innovative, come
quelle che l'Italia sta cercando di provare. Abbiamo fatto passaggi in avanti al
livello dell'Unione Europea, dove stiamo lavorando per dare una interpretazione
più moderna delle convenzioni che abbiamo, ma le abbiamo scritte molti anni fa.
E al livello dell'EPC, con Cure, abbiamo promulgato un'altra declarazione, con
la proposta che abbiamo fatto con Cure, la posizione è abbastanza chiarissima,
dobbiamo combattere contro i trafici, dobbiamo evitare che una nuova crisi, come
quella che abbiamo in 2015, passi di nuovo, e dobbiamo cooperare con i paesi di
origine e transito. E questo mi porti alla prima e alla seconda parte della
domanda che ha fatto Nikos. Il problema è se abbiamo questa crisi poli, possiamo
affrontare la crisi in modo che sia aderente? La mia idea è che il livello
dell'Unione Europea, per esempio, in anni recenti, abbiamo mostrato la nostra
capacità di rispondere alle emergenze. Abbiamo fatto questo con la crisi della
pandemia, abbiamo fatto questo con l'Ukraine, ma penso anche che ora dobbiamo
stoppare, passare dalla nostra capacità di reagire, alla nostra capacità di
anticipare. Questo significa che dobbiamo concentrarci molto più sulla nostra
strategia nel termine lungo e queste discussioni sono molto utili per questo. La
nostra strategia, in mia opinione, deve essere più forte, non solo sui paesi
similari, ovviamente è molto importante, ma anche nella nostra città geografica.
Ovviamente ciò che stiamo facendo ora con i paesi similari, con l'EPC, è molto
importante e dobbiamo migliorare e sono molto felice che Mark sia qui. Ma penso
che dovremmo anche concentrarci, per esempio dal punto di vista dell'Italia,
quindi penso che dovremmo concentrarci molto più, per esempio, sulla nostra
città mediterranea. Penso anche che dovremmo lavorare su un forum come l'EPC,
che è fondamentale, ovviamente per noi in questi tempi, concentrati nella città
geografica del mediterraneo. Nel passato con l'Urzula, con gli altri Stati
membri, con l'Unione Europea, abbiamo provato a trattare la migrazione, siamo
stati forzati a pagare molto più attenzione ai paesi attraverso il Mediterraneo.
Ora, in mia opinione, il rischio è di ampliare questa approach, di portare
insieme energia di rivolto di sicurezza e sviluppo nella nostra cooperazione con
i paesi nello spazio. Per questo è la migliore forma in cui dobbiamo rispondere
a una crisi che non è solo una crisi. Ci sono molte crisi insieme, ma
infortunatamente sono parte della stessa crisi e dobbiamo trattare tutto.
Grazie.",low,medium,low,none

#---------end of csv-----------------#

the above speech was classified with this prompt:
"""
 i will paste in this chat speeches from italian prime minister formatted as
csv. you have to classify the "text" coloumn.


for each speech, classify them.

"hate_speech", "negativity", and "aggressiveness" are classified with

three levels, respectively "low", "mid" and "high". each speech has to

be classfied with one of the three values for each .


"target", on the other hand, should be classified independently from the

previous three. what i want is just to know whether the speech have one
```

```
of the following five targets: "none" if there is no target, "pol_adv"
if it speaks about political adversaries, "minor_etn" if it speaks about
ethnical minorities (eg balck people), "minor_gnd" if it speaks about
gender minorities (e.g., women or LGBT community), and "minor_rel" if it
speaks about religious miniorities (e.g., muslims).

give me the results in the form: politician, date ,low,low,low,pol_adv
"""

#--------------------#
unfortunately, most of my speeches have very low diversity in their
classification, meaning that data is heavily skewred towards low hate speech,
low aggressiveness, low negativity and target = none.
make me a syntetic csv dataset with 100 speeches, where: politician="synthetic";
random historical date; location = "UNKNOWN"; no title, no url, no audio file
coloumns.
all speeches on the other hand must have hate speech, aggressiveness, and
negativity at least medium for all three of them. try to make it so that the
speeches have different amount of those so that there is as much diverisity as
possibile.
Try to make it so that all speeches have one of the possible four targets:
"pol_adv" if it speaks about political adversaries, "minor_etn" if it speaks
about ethnical minorities (eg balck people), "minor_gnd" if it speaks about
gender minorities (e.g., women or LGBT community), and "minor_rel" if it speaks
about religious miniorities (e.g., muslims).
when you make a speech, classify it too.
to reiterate the dataset coloums, they are the following:
[politician,historical_date,location,text,hate_speech,negativity,
aggressiveness,target]
```

Listing 3: Prompt to generate 100 synthetic speeches with high hostility and diverse targets. The speeches were generated with Gemini 3.6 Flash.

1.5. LLM's Prompt To Annotate Dataset

```
RUBRIC_HEADER = """Sei un annotatore esperto di scienze politiche e linguistica
computazionale.
Il tuo compito è classificare un discorso pronunciato da un Presidente del
Consiglio italiano."""

# think=True group: hate_speech + target
SYSTEM_PROMPT_THINK = RUBRIC_HEADER + """

---
### DIMENSIONI E CATEGORIE

1) hate_speech: presenza di incitamento all'odio verso individui o gruppi
protetti.
   - low: assente o non significativa.
```

```
   - mid: presente in forma velata, allusiva, tramite "dog whistles" o pregiudizi impliciti.
   - high: presente in forma esplicita, deumanizzante o aperta ostilità sistematica.

2) target: il bersaglio principale di eventuale ostilità/critica nel discorso (assegna il target
   anche se il discorso è nel complesso pacato, identificando verso chi è orientato).
   - none: nessun bersaglio specifico identificabile.
   - pol_adv: avversari politici, opposizioni, partiti, burocrazia europea o altre istituzioni.
   - minor_etn: minoranze etniche, persone straniere, migranti di specifiche etnie.
   - minor_gnd: minoranze di genere, donne, comunità LGBTQ+.
   - minor_rel: minoranze religiose (es. musulmani, ebrei).

---
### ESEMPI GUIDA PER LA CALIBRAZIONE

#### Esempi per `hate_speech`:
- **low**:
  - "Dobbiamo investire nella scuola pubblica e garantire il salario minimo per sostenere le famiglie in difficoltà."
  - "La transizione ecologica richiede una visione industriale chiara che tuteli l'occupazione e le piccole imprese."
  - "Rafforzeremo il presidio del territorio e la cooperazione tra le forze dell'ordine e le comunità locali."
- **mid**:
  - "Non possiamo permettere che i valori della nostra civiltà vengano diluiti da chi arriva da fuori senza alcuna intenzione di assimilarsi."
  - "Certe culture tradizionaliste restano intrinsecamente incompatibili con la tutela dei diritti fondamentali e dello Stato di diritto."
  - "I quartieri storici stanno perdendo la propria identità a causa di una presenza straniera ormai dominante che rifiuta le nostre regole."
- **high**:
  - "L'immigrazione è un'invasione pianificata che porterà degrado e criminalità nelle nostre città. Bisogna eliminarla sistematicamente strada per strada, casa per casa."
  - "Questi gruppi parassitari infestano la nostra società e vanno estirpati prima che distruggano definitivamente la nostra razza e la nostra storia."
  - "Alcune minoranze sono portatrici biologiche di violenza e inciviltà: vanno cacciate con ogni mezzo dal nostro suolo patrio."

#### Esempi per `target`:
- **none**:
  - "Oggi approviamo la riforma del codice della strada per ridurre gli incidenti e tutelare i giovani."
  - "Il piano di digitalizzazione della pubblica amministrazione consentirà di ridurre radicalmente i tempi di attesa."
  - "I finanziamenti per la prevenzione del dissesto idrogeologico copriranno tutte le regioni a rischio."
- **pol_adv**:
  - "La precedente maggioranza ha lasciato buchi di bilancio incalcolabili per fare propaganda elettorale."
  - "L'Unione Europea pretende di imporre direttive ideologiche che penalizzano le nostre filiere produttive nazionali."
  - "La magistratura politicamente orientata continua a travalicare i propri confini costituzionali per ostacolare l'esecutivo."
- **minor_etn**:
```

```
  - "È necessario bloccare le partenze e combattere le reti di clandestinità che destabilizzano la sicurezza nazionale."
  - "Certi flussi migratori incontrollati provenienti dall'Africa subsahariana importano modelli criminali inaccettabili."
  - "Non tollereremo zone di franchigia gestite da bande etniche straniere nelle periferie dei nostri capoluoghi."
- **minor_gnd**:
  - "Ci opporremo a chi vuole scardinare la famiglia naturale imponendo teorie ideologiche nelle scuole."
  - "La propaganda sull'identità di genere mira a cancellare la figura della madre e il ruolo biologico della donna."
  - "I diritti delle donne vengono calpestati dall'ossessione per il politicamente corretto e la fluidità di genere."
- **minor_rel**:
  - "Alcune comunità religiose pretendono di applicare le proprie leggi teocratiche sul nostro territorio nazionale."
  - "Il proliferare di centri di preghiera abusivi legati all'Islam radicale rappresenta una minaccia diretta ai nostri valori laici."
  - "Non faremo concessioni a chi usa il proprio culto per giustificare la sottomissione femminile e l'odio verso l'Occidente."

---
### FORMATO DI OUTPUT

Rispondi ESCLUSIVAMENTE con un oggetto JSON con queste due chiavi (hate_speech, target),
senza testo aggiuntivo, markdown o spiegazioni."""

# think=False group: negativity + aggressiveness
SYSTEM_PROMPT_NOTHINK = RUBRIC_HEADER + """

---
### DIMENSIONI E CATEGORIE

1) negativity: tono complessivo negativo del discorso (critica, pessimismo, minacce percepite, allarmismo).
   - low: tono prevalentemente neutro, costruttivo o positivo.
   - mid: presenza moderata di toni critici o allarmi su specifici problemi nazionali.
   - high: tono fortemente cupo, allarmista, focalizzato su crisi imminenti o declino.

2) aggressiveness: aggressività verbale, toni polemici o ostili verso interlocutori/avversari.
   - low: tono pacato, formale, istituzionale.
   - mid: presenza moderata di polemica, velate stoccate o sarcasmo istituzionale.
   - high: tono marcatamente polemico, attacchi diretti e personali, linguaggio di scontro.

---
### ESEMPI GUIDA PER LA CALIBRAZIONE

#### Esempi per `negativity`:
- **low**:
  - "I dati sull'occupazione sono incoraggianti e la crescita economica dimostra la resilienza del nostro tessuto produttivo."
  - "La riforma della giustizia procede spedita, restituendo efficienza e tempi certi a cittadini e imprese."
  - "Il nostro posizionamento internazionale si rafforza grazie a nuove partnership strategiche sull'energia."
```

```
- **mid**:
  - "I sistemi sanitario e scolastico presentano problematiche infrastrutturali apparentemente insanabili."
  - "L'instabilità geopolitica globale proietta ombre preoccupanti sull'approvvigionamento delle nostre materie prime."
  - "La burocrazia soffocante e i tempi della giustizia civile continuano a frenare gli investimenti esteri nel Paese."
- **high**:
  - "Il paese è sull'orlo del baratro finanziario; ereditiamo un disastro sistemico che rischia di spazzare via i risparmi degli italiani."
  - "Siamo di fronte a un declino demografico e sociale irreversibile che sta portando la Nazione alla completa estinzione."
  - "Il collasso della sicurezza urbana ha trasformato le nostre metropoli in zone di guerra allo sbaraglio."

#### Esempi per `aggressiveness`:
- **low**:
  - "Accogliamo le osservazioni delle opposizioni nel merito, tuttavia il Governo continuerà con la linea dettata dagli elettori."
  - "Il confronto parlamentare è la sede naturale per affinare i testi di legge nell'interesse generale."
  - "Valuteremo con attenzione tutti gli emendamenti proposti dalle forze di minoranza durante il percorso in commissione."
- **mid**:
  - "Riconosco il diritto dell'opposizione di protestare, anche se la loro memoria storica appare alquanto corta e di comodo."
  - "Ci danno lezioni di rigore di bilancio gli stessi banchieri e tecnocrati che hanno affossato i conti pubblici nel decennio scorso."
  - "Spiace constatare come parte dei media preferisca la polemica strumentale all'analisi obiettiva dei fatti."
- **high**:
  - "Dall'opposizione arrivano solo menzogne sfrontate e sciacallaggio politico da parte di chi ha svenduto la nazione per anni."
  - "Siete dei traditori del popolo italiano, dei cospiratori servili che prendono ordini da potenze e burocrazie straniere."
  - "La vostra ipocrisia fa schifo: avete le mani sporche di sangue per le politiche criminali che avete approvato!"

---
### FORMATO DI OUTPUT

Rispondi ESCLUSIVAMENTE con un oggetto JSON con queste due chiavi (negativity, aggressiveness),
senza testo aggiuntivo, markdown o spiegazioni."""

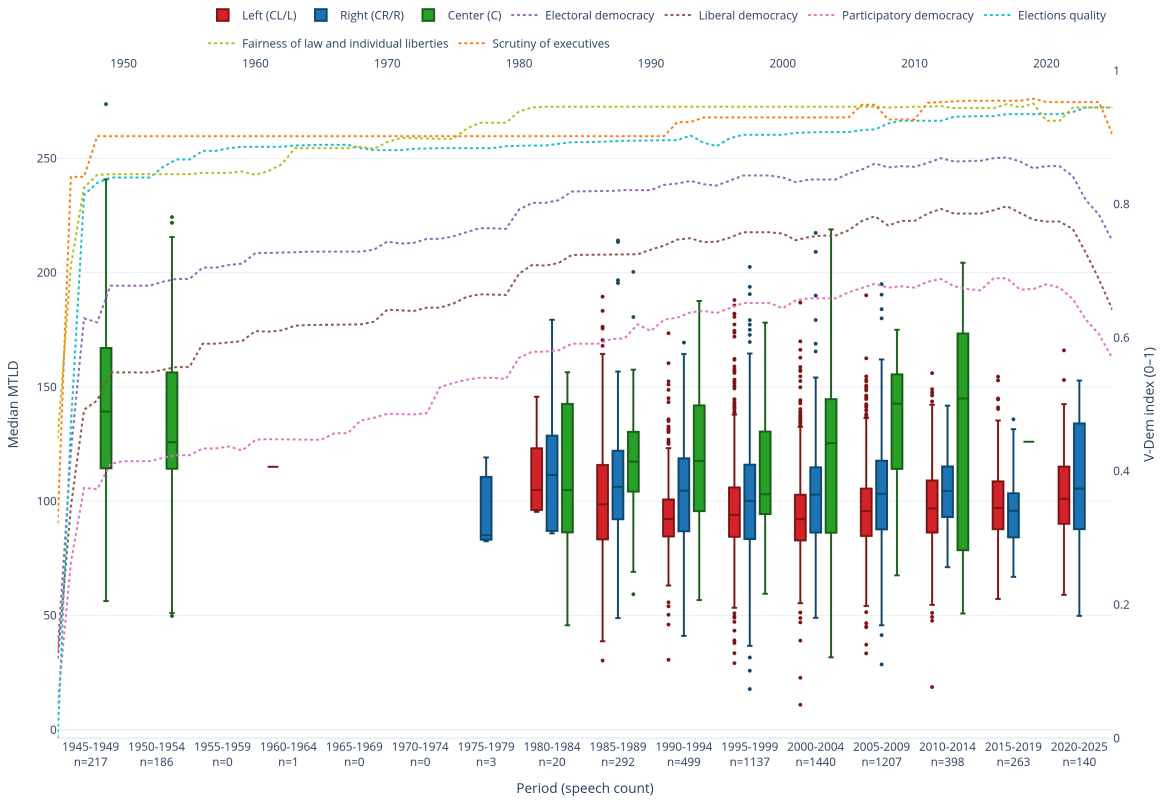

JSON_SCHEMA_THINK = {
    "type": "object",
    "properties": {
        "hate_speech": {"type": "string", "enum": LEVEL_VALUES},
        "target": {"type": "string", "enum": TARGET_VALUES},
    },
    "required": THINK_COLS,
}

JSON_SCHEMA_NOTHINK = {
    "type": "object",
    "properties": {
        "negativity": {"type": "string", "enum": LEVEL_VALUES},
        "aggressiveness": {"type": "string", "enum": LEVEL_VALUES},
    },
```

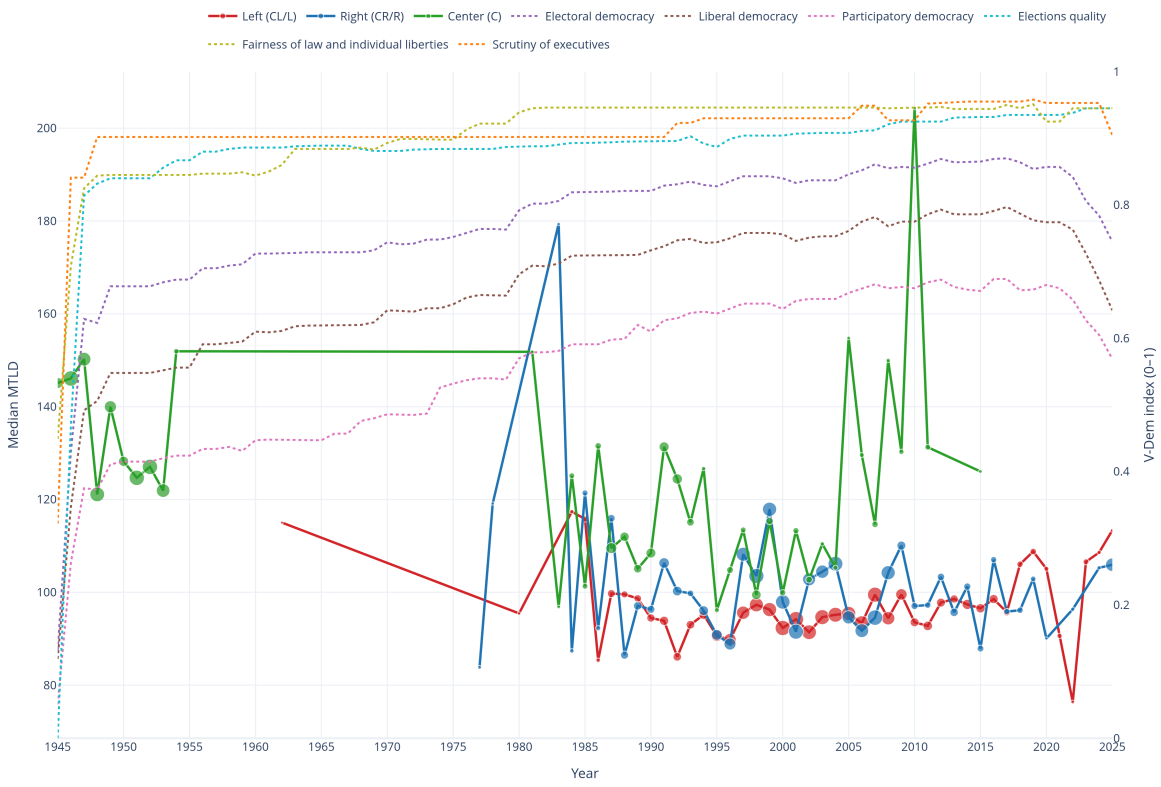
```
    "required": NOTHINK_COLS,
}
```

Listing 4: Prompts to be fed into Gemma 4 to classify the corpus and their respective JSON output schemas. One prompt to classify negativity and aggressiveness with thinking disabled , and one prompt to classify hate speech and target with thinking enabled.

1.6. V-Dem MTLD Evolution Graphs



(a) MTLD values in a Boxplot by 5-year periods, with VDEM indices. As above, each grey box combines all speeches for all political leanings in that 5-year period, while the colored boxes represent respectively center-leaning (in green), left-leaning (in red), and right-leaning (in blue) MTLD values in that 5-year preiod.



(b) MTLD values plotted for each year, aggregating speeches from three political leanings, respectively colored as green (center), red (left), and blue (right). Each dot of the graph (for each year) is scaled based on the amount of speeches available for that specific year. The MTLD line chart is plotted against VDEM indices.

Figure 12: MTLD values aggregated for each year, plotted against VDEM indices. There does not seem to be a correlation between a decrease in complexity and a decrease in democratic indices. Instead, the graph seems to suggest the opposite. There seem to be a reduction in MTLD variance starting from 1989 all the way to 2019, whith an abrupt inversion in the 2020-2025 period. If this is a correlation with the inflection of some VDEM indices, it is not clear.